# CSC432 – Information Security

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Lecture No 3

# Cryptography and Encryption

# Cryptography

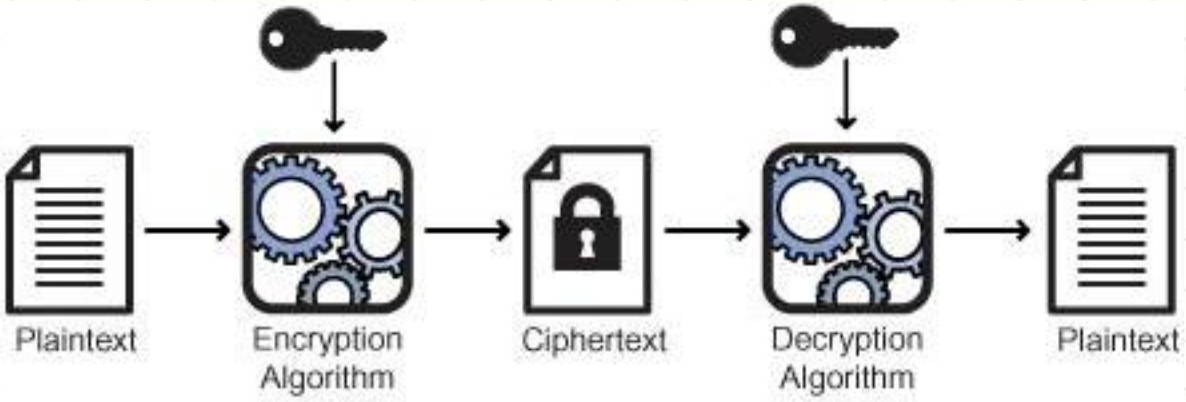
- The word Cryptography comes from the Greek words, "Crypt" means (hidden or secret) and "Graphy" means (writing)
- Cryptography = 'the art of secret writing'
- The basic service provided by cryptography is the ability to send information between participants in a way that prevents others from reading it

# Encryption

- Alternatively, Encryption is the actual process of transforming information into an illegible format
- Encryption basically is some process or algorithm to make information hidden or secret
- To make that process useful, you need some code to make information accessible
- Modern day encryption uses different types of algorithms to achieve results that vary in complexity

# Encryption

- A message in its original form is known as **plaintext** or cleartext
- The mangled information is known as **ciphertext**
- **Encryption** is a process by which a message (called plaintext) is transformed into another message (called ciphertext) using a mathematical function and a special encryption password (called a key)
- The reverse of encryption is called **decryption**



# Cryptography

- **Cryptographic systems** tend to involve both an algorithm and a secret value (means they use encryption/decryption)
- While **cryptographers** invent clever secret codes, **cryptanalysts** attempt to break these codes
- These two disciplines constantly try to keep ahead of each other

# Cryptography

- Cryptographic algorithms involve substituting one thing for another, in many possible ways
- A cipher is an algorithm for performing encryption or decryption — a series of well-defined steps that can be followed as a procedure
- Example: Caesar cipher
  - substitute each letter by the letter that appears k letters later in the alphabet; while producing what looks like gibberish
  - there are only 25 possible keys available

# Cryptography

Q:  How hard is it to break these simple ciphers?

Objective is to recover key not just message

- **Brute force attack (attempt all possibilities)**
- Simple with the Caesar cipher, but gets quite difficult with monoalphabetic or polyalphabetic ciphers

- **Cryptanalytic attack**
- Ciphertext-only attack:  use statistics and other information to decrypt intercepted ciphertext
  - For example, simple statistics on letter placement and occurrence in English makes further decryption simpler

# Cryptography

- <u>Known-plaintext attack</u>: if some of the plaintext is known, one could uncover some of the plaintext-ciphertext mappings, making decryption easier
  - For example, if we knew “Alice” was in the message, we get some of the mappings right away without much difficulty

- <u>Chosen-plaintext attack</u>: the intruder can choose the plaintext (to be encrypted) message and receive the (corresponding) ciphertext form

<u>It gets much easier to decrypt messages; monoalphabetic ciphers are easily broken, and polyalphabetic ciphers get much easier too</u>

# Cryptography

- An encryption scheme is computationally secure if
  - The cost of breaking the cipher exceeds the value of information
  - The time required to break the cipher exceeds the lifetime of information

  - Example: Encrypt = division
  - 433
  - 48 R 1 (using divisor of 9)

Unconditionally secure - algorithm is secure without any condition (impossible)
Provably secure - it is proved till yet that algorithm is secure but can be cracked in future
Computationally secure - Time after which algorithm break, information is of no worth. Like if an algorithm take 2 days to decrypt text, and after this time information is not usable, it is secure

Unconditionally secure - Provably secure - Computationally secure

# Cryptography vs Steganography

- Cryptography is the practice of 'scrambling' messages so that even if detected, they are very difficult to decipher
- Steganography (means concealed writing) is to conceal the message such that the very existence of the hidden is 'camouflaged'

- Steganography is data inside data
- It is mostly used to hide text inside pictures or sound files (digital watermarking)

# Types of Cryptography

- **Symmetric key cryptography**: encryption and decryption keys are identical, so the key must be kept secret
- This approach is also called secret/private key cryptography

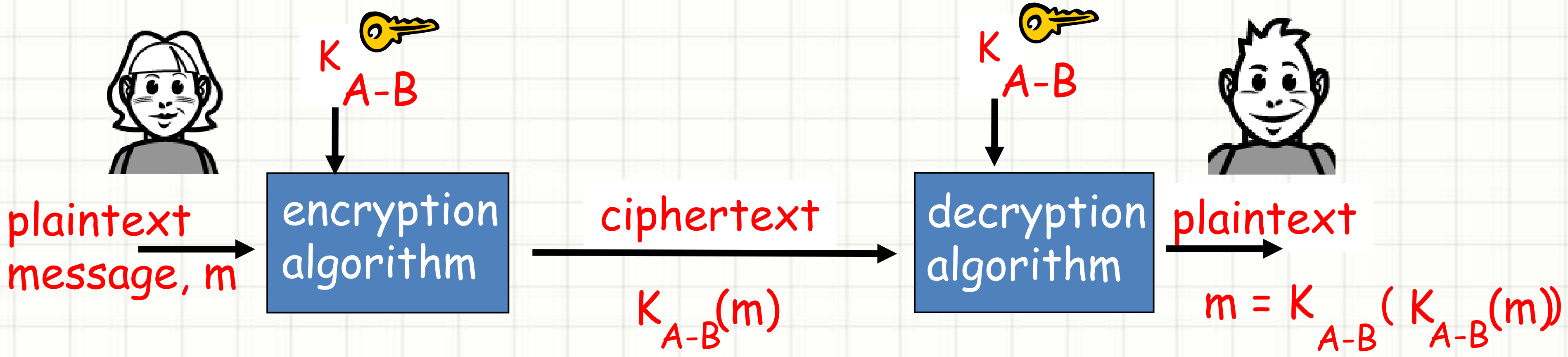
- **Asymmetric key cryptography**: different keys for encryption and decryption (one public, the other private)
- This approach is also called two key/public key cryptography

# Symmetric Key Cryptography

- Same key decrypts and encrypts information
- The encryption functions used need not be secret, but the keys used must be secret

- Examples:
  - ROT13: Very simple rotation algorithm
  - **Caesar cipher: Another (better) rotation algorithm**
  - Crypt: Original Unix encryption program
  - **DES: Data Encryption Standard**
  - **AES: Advanced Encryption Standard**
  - **IDEA: International Data Encryption Algorithm**
  - Skipjack: U.S. National Security Agency developed algorithm

# Symmetric Key Cryptography




- Bob and Alice share the same (symmetric) key: $K_{A-B}$
- For example, the key is knowing substitution pattern in a cipher

# Symmetric Key Cryptography

Key Issues in Symmetric Key Cryptography

- Question: How do Bob and Alice agree on key value? What if Bob and Alice have never "met" before?
- Even Better Question: How is the agreed upon key distributed to both Bob and Alice in a secure fashion?

# Asymmetric Key Cryptography

- Radically different approach (two different keys)
- Sender and receiver do not share secret key
- Public encryption key known to all
- Private decryption key known only by the owner

- Examples
  - **Diffie-Hellman: the first public key approach proposed**
  - **RSA: the best known public key system, developed by Rivest, Shamir, and Adleman**
  - DSA: Digital Signature Algorithm, developed by the U.S. National Security Agency (NSA)

# Asymmetric Key Cryptography

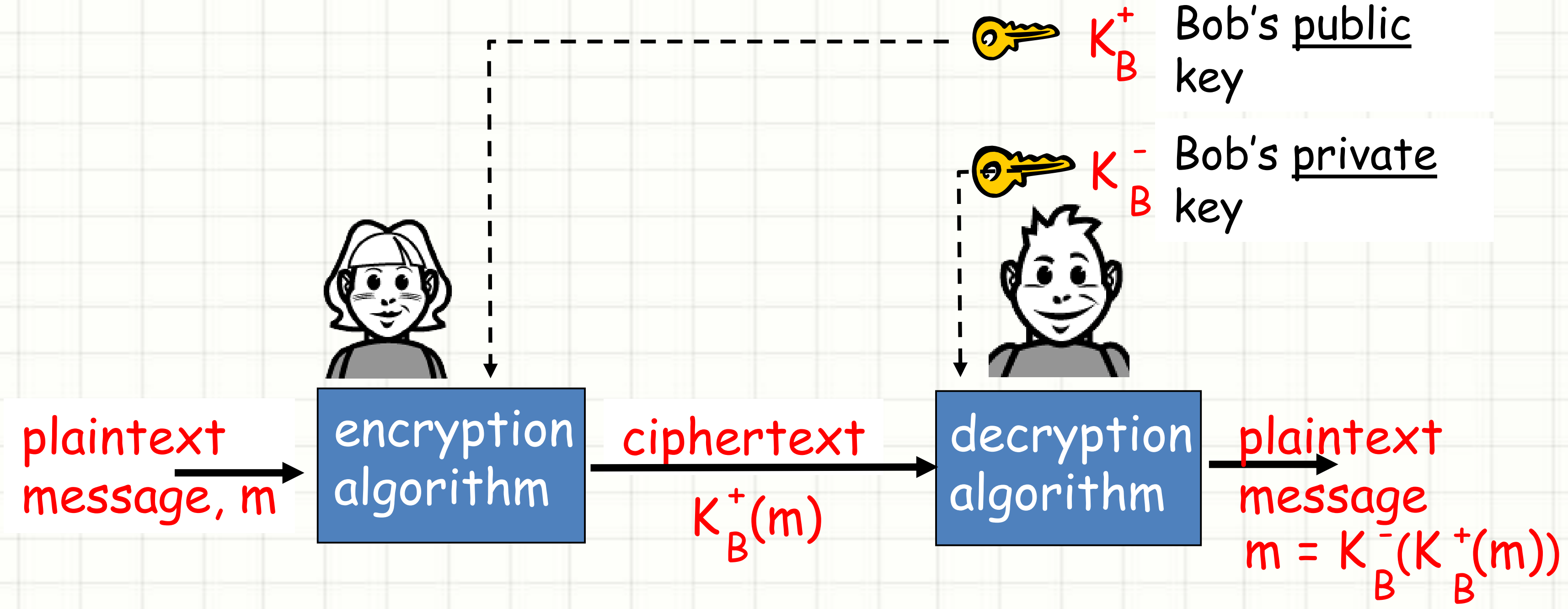
Keys are generated in pairs

- Public key is publicly registered so everyone knows it, and private one is kept secret by the owner
- Each key can decrypt what the other encrypts, but not what it encrypts itself (that why asymmetric)

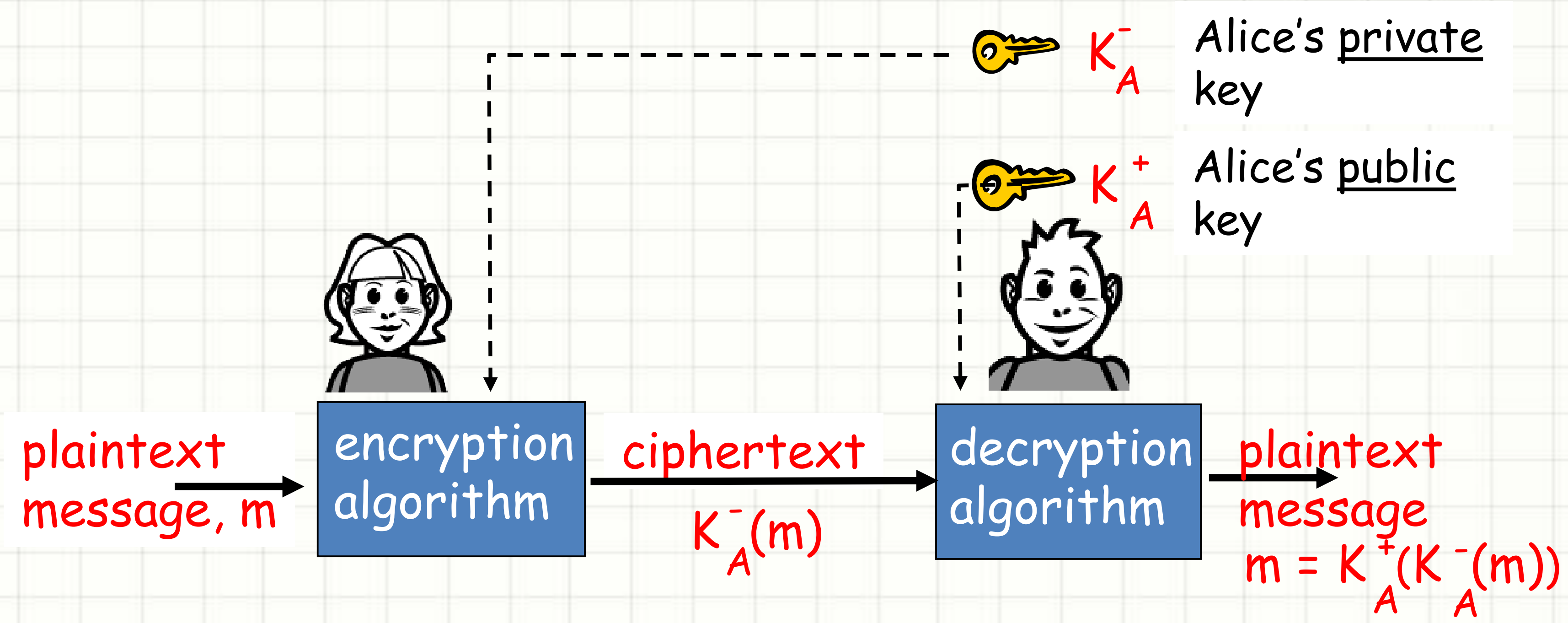
Important properties of key generation:

- There is a one-to-one correspondence in the generated key pairs – if one key can decrypt a message, it must have been encrypted by the other
- It must be extremely difficult, if not impossible, to deduce the private key when given a public key

# Asymmetric Key Cryptography



# Asymmetric Key Cryptography



# Asymmetric Key Cryptography

- This property will be very useful later:

$$K_B^-(K_B^+(m)) = m = K_A^+(K_A^-(m))$$

- use private key first, followed by public key
- use public key first, followed by private key

- Result is the same

# Applications of Cryptography

- Transmitting over an insecure channel
- Secure storage on insecure media

- Everything that SKC does can be done by PKC and Digital Signatures

# Cryptography Terms Summary

- Plaintext - original message
- Ciphertext - coded message
- Cipher - algorithm for transforming plaintext to ciphertext
- Key - info used in cipher known only to sender/receiver
- Encipher (encrypt) - converting plaintext to ciphertext
- Decipher (decrypt) - recovering plaintext from ciphertext
- Cryptography - study of encryption principles/methods
- Cryptanalysis (codebreaking) - study of principles/ methods of deciphering ciphertext without knowing key
- Cryptology - field of both cryptography and cryptanalysis

# Ciphers and Their Types

- **Classical Ciphers**
  - Transposition Cipher
  - Substitution Cipher
    - Monoalphabetic
    - Polyalphabetic

- **Modern Cipher**
  - Asymmetric
  - Symmetric
    - Block
    - Stream

# Ciphers and Their Types

- **Transposition**: rearrange bits or characters in the data (permutation)
- **Substitution**: replace bits, characters, or blocks of characters with substitutes
  - **Monoalphabetic**: A single alphabet is used to encrypt the entire plaintext message
  - **Polyalphabetic**: A more complex substitution that uses a different alphabet to encrypt each bit, character, or character block of a plaintext message
- A **block cipher** is one that breaks a message up into chunks and combines a key with each chunk
- A **stream cipher** is one that applies a key to each bit, one at a time

# Ciphers and Their Types

**Caesar Cipher**

- One of the simplest examples of a cipher is the Caesar cipher
- It is a type of substitution cipher in which each letter in the plaintext is replaced by a letter some fixed number of positions down the alphabet
- For example, with a shift of 3, 'A' would be replaced by 'D', 'B' would become 'E', and so on

Plaintext: the quick brown fox jumps over the lazy dog

Ciphertext: WKH TXLFN EURZQ IRA MXPSV RYHU WKH ODCB GRJ

# Ciphers and Their Types

## Vigenère Cipher

- The most common polyalphabetic cipher
- Vigenère cipher starts with a 26 x 26 matrix of alphabets in sequence
- First row/column starts with 'A', second row/column starts with 'B', etc
- It requires a keyword that the sender and receiver know ahead of time
- Each character of the message is combined with the characters of the keyword to find the ciphertext character

# Ciphers and Their Types

Vigenère Cipher Matrix

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| B | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A |
| C | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B |
| D | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C |
| E | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D |
| F | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E |
| G | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F |
| H | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G |
| I | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H |
| J | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I |
| K | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J |
| L | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K |
| M | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L |
| N | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M |
| O | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
| P | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O |
| Q | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
| R | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
| S | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R |
| T | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S |
| U | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
| V | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |
| W | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V |
| X | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | X |
| Y | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | X | Y |
| Z | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | X | Y | Z |

# Ciphers and Their Types

Vigenère Cipher Example

During encryption - look plain text in column, while during decryption, look key in column and first letter in the column

- Message = SEE ME IN MALL
- Keyword = INFOSEC

| Encryption | Decryption |
|---|---|
| S E E M E I N M A L L | A R J A W M P U N Q Z |
| I N F O S E C I N F O | I N F O S E C I N F O |
| ------------------------------ | ------------------------------ |
| A R J A W M P U N Q Z | S E E M E I N M A L L |

- As its polyalphabetic, same plaintext character is substituted by different ciphertext

# Ciphers and Their Types

## Playfair Cipher

- The best known multiple-letter cipher (polyalphabetic), which treats digrams in the plaintext as single units
- It is based on a 5x5 matrix of letters constructed using a keyword
- Suppose we have a keyword "monarchy"
- The matrix is constructed by filling in the letter of the keyword (minus duplicate) from left to right and from top to bottom in alphabetic order
- The letter I and J count as one letter (as we have max 25 spaces available)

# Ciphers and Their Types

## Playfair Cipher Matrix

- Keyword = "monarchy"

| | | | | |
|---|---|---|---|---|
| m | o | n | a | r |
| c | h | y | b | d |
| e | f | g | i/j | k |
| l | p | q | s | t |
| u | v | w | x | z |

# Ciphers and Their Types

Playfair Cipher Example

- Plaintext is encrypted two letters at a time, according to the following rules;
- Repeating plaintext letters that would fall in the same pair are separated with a filler letter, such as x, so that 'balloon' would be treated as 'ba lx lo on'
- Plaintext letters that fall in the same row of the matrix are each replaced by the letter to the right, with the first element of the row circularly following the last. For example, ar is encrypted as RM
- Plaintext letters that fall in the same column are replaced by the letter beneath, with the top element of the row circularly following the last. For example, mu is encrypted as CM

# Ciphers and Their Types

- Otherwise, each plaintext letter is replaced by the letter that lies in its row and the column occupied by the other plaintext letter. Thus, hs becomes BP and ea becomes IM (or JM)

Example:

Plaintext: see me tomorrow

se em et om or ro wx

Ciphertext: li lc kl no nm mn xz

- Decryption is just the reverse of encryption

# Ciphers and Their Types

## Affine Cipher

- The affine cipher is a monoalphabetic substitution cipher
- The encryption/decryption process is substantially mathematical

- Step one: Substitute each letter in your plaintext message with a number (range 0 to m-1)

| Plaintext Alphabet | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Plaintext Value | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

# Ciphers and Their Types

## Affine Cipher Example

- Create a permutation of the alphabet by replacing each a with the result of a simple equation:
- E(x) = (ax + b) mod m

- m is 26 in this case as total number of alphabets in English language
- a is relatively prime to 26 (or the length of whatever alphabet you're using), and b is an arbitrary integer of your choice
- a and b need to be known to decrypt

a is relatively prime to 26 means that their common factor must not be other than '1'

# Ciphers and Their Types

## Affine Cipher Example

- lets encrypt the plaintext "affine cipher", using the key a = 5, b = 8

| Plaintext | a | f | f | i | n | e | c | i | p | h | e | r |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| x | 0 | 5 | 5 | 8 | 13 | 4 | 2 | 8 | 15 | 7 | 4 | 17 |
| 5x+8 | 8 | 33 | 33 | 48 | 73 | 28 | 18 | 48 | 83 | 43 | 28 | 93 |
| (5x+8) mod 26 | 8 | 7 | 7 | 22 | 21 | 2 | 18 | 22 | 5 | 17 | 2 | 15 |
| Ciphertext | I | H | H | W | V | C | S | W | F | R | C | P |

# Ciphers and Their Types

Affine Cipher Example

- In decryption, we must perform the opposite (or inverse) functions on the ciphertext to retrieve the plaintext, using the following equation;
- D(x) = c(x - b) mod m

- c is the modular multiplicative inverse of a i.e., a*c = 1 mod m
- We know that, a = 5, b = 8
- The first step here is to find the inverse of a, which in this case is 21
- (since 21 x 5 = 105 = 1 mod 26, as 26 x 4 = 104, and 105 - 104 = 1)

choose 'c' such that if we multiply it with 'a' and then divide by 26, remainder must be '1'

# Ciphers and Their Types

## Affine Cipher Example

- lets do decryption of the ciphertext

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ciphertext | I | H | H | W | V | C | S | W | F | R | C | P |
| y | 8 | 7 | 7 | 22 | 21 | 2 | 18 | 22 | 5 | 17 | 2 | 15 |
| 21(y-8) | 0 | -21 | -21 | 294 | 273 | -126 | 210 | 294 | -63 | 189 | -126 | 147 |
| 21(y-8) mod 26 | 0 | 5 | 5 | 8 | 13 | 4 | 2 | 8 | 15 | 7 | 4 | 17 |
| Plaintext | A | F | F | I | N | E | C | I | P | H | E | R |

-n mod m = m - a

# Ciphers and Their Types

## Rail Fence Cipher

- A transposition cipher that gets its name from the way in which it is encoded
- Plaintext is written downwards on successive "rails" of an imaginary fence, then moving up when we get to the bottom
- Encrypted message (cipertext) is then read off in rows

# Ciphers and Their Types

Rail Fence Cipher Example

- Using three "rails" and a plaintext 'WE ARE  DISCOVERED. FLEE AT ONCE'

| | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| W | . | . | . | E | . | . | . | C | . | . | . | R | . | . | . | L | . | . | . | T | . | . | . | E |
| . | E | . | R | . | D | . | S | . | O | . | E | . | E | . | F | . | E | . | A | . | O | . | C | . |
| . | . | A | . | . | . | I | . | . | . | V | . | . | . | D | . | . | . | E | . | . | . | N | . | . |

- Ciphertext = WECRL TEERD SOEEF EAOCA IVDEN

- Decryption is just the reverse of encryption

# Ciphers and Their Types

## One-time Pad

- Use a random key that is as long as the message so that the key need not be repeated
- The key is used to encrypt and decrypt a single message, and then is discarded
- Perfectly secure, unbreakable (if used correctly) because it produces random output (from the random key) that bears no statistical relationship to the plaintext
- Drawbacks: large quantities of random keys needed, key distribution and protection (both sender and receiver)

# Ciphers and Their Types

One-time Pad Example

plaintext: SECRETMESSAGE
one-time pad: CIJTHUUHMLFRU
ciphertext: UMLKLNGLEDFXY

# References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THANKS